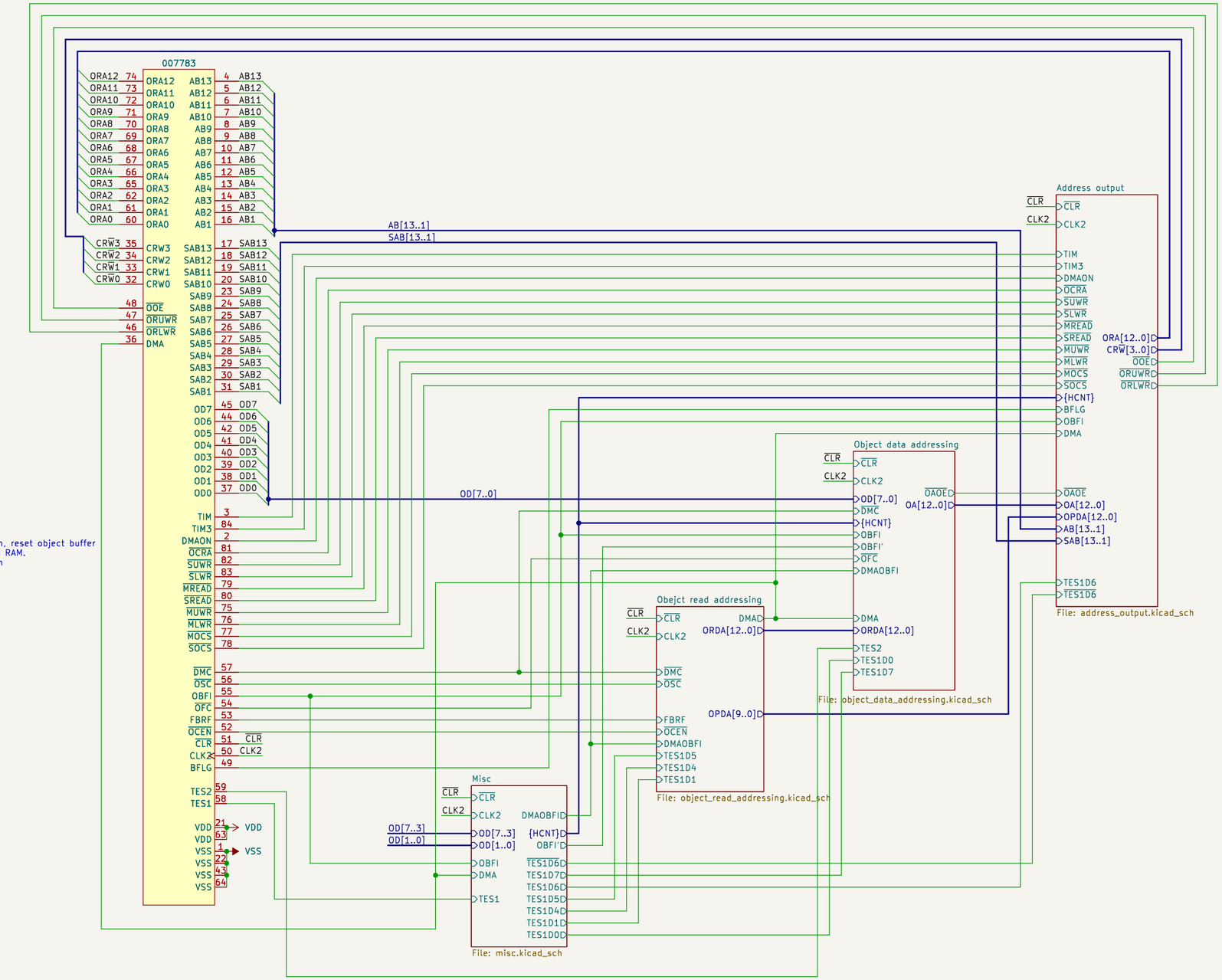
1. OBFI: (OBJBUFINIT) zeros are written to 1800h..1AFFh.
2. DMA: OA[12:0] active.
   a) 8 words are read from addresses 0000h..0007h, 0028h..002Fh,...,17E0h..17E7h
      by the sprite data parser 007784 and if the sprite is enabled with BEFLAG
      they are written to the buffer OBJBUF.

OBJRAM
0000h..17E7h Unprocessed data
1800h..1BFFh Processed data

WD0	WD1	WD2	WD3	WD4	WD5	WD6	WD7	WD0	WD1	WD2	WD3
0000	0001	0002	0003	0004	0005	0006	0007	BA1	BA2	BA3	BA4

OBJBUF, Buffer Address
BA1 : OA[12:0] = 1800h + OD''[7:0] << 2
BA2 : OA[12:0] = BA1 + 1
BA3 : OA[12:0] = BA1 + 2
BA4 : OA[12:0] = BA1 + 3

OBFI	H8	OA	OAOE	
1	X	{3'b110,OWRA[9:0]}	1	Writes to 1800h..1BFFh, reset object buffer
0	0	ORDA[12:0]	0	Read data from object RAM.
0	1	{3'b110,OD''[7:0],OWRA[2:1]}	1	Writes to 1800h..1BFFh



Ulf Skutnabba, twitter: @skutis77

Sheet: /
File: 007783.kicad_sch

Title: Konami 007783

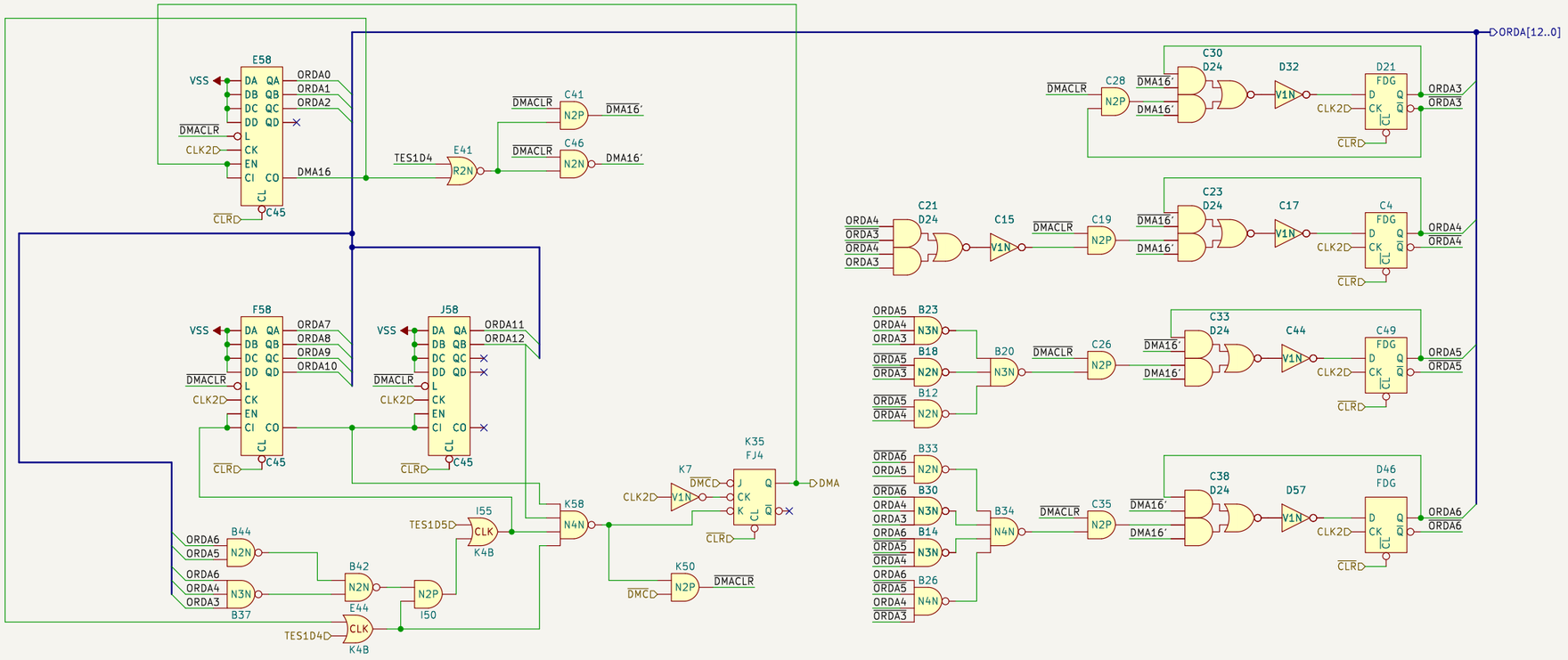
Size: A3	Date: 2023-10-10	Rev:
KiCad E.D.A. 8.0.5		Id: 1/5

Object Buffer Write Address

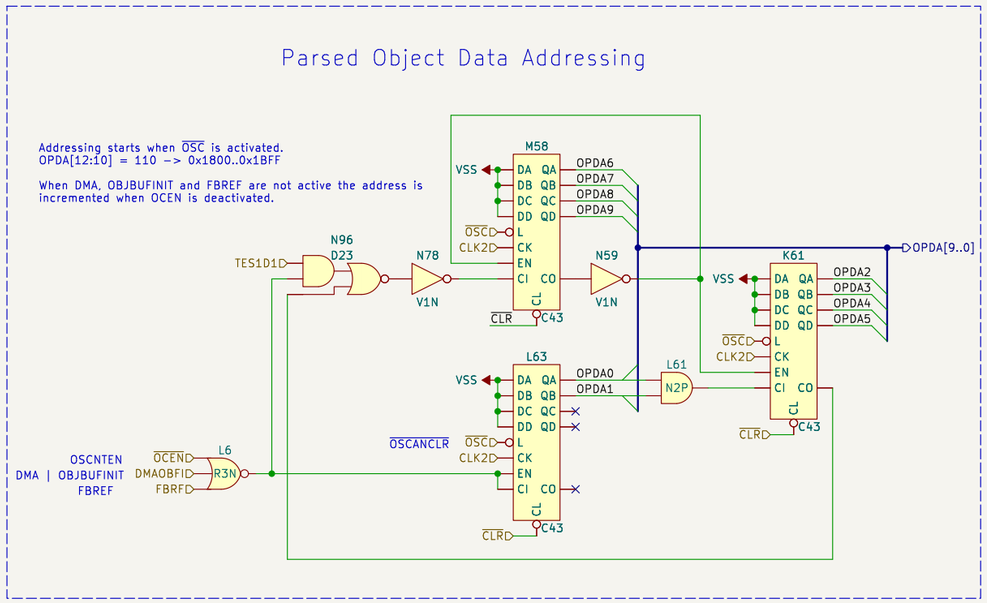
OBFI = 1–> H8OBFI = 1, $\overline{H8OBFI}$ = 0, $\overline{OAOE}$ = 1, Only writes.
OBFI = 0 –> H8OBFI = H8, $\overline{H8OBFI}$ = H8, $\overline{OAOE}$ = H8

Sheet: /Object data addressing/		
File: object_data_addressing.kicad_sch		
Title: Konami 007783		
Size: A3	Date: 2023-10-10	**Rev:**
KiCad E.D.A. 8.0.5		Id: 2/5

Object Read Addressing



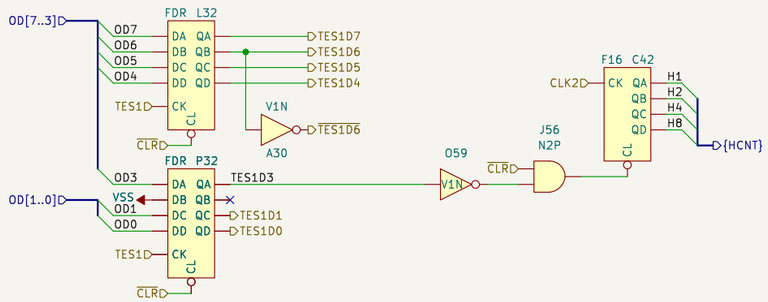
Parsed Object Data Addressing



Sheet: /Obejct read addressing/		
File: object_read_addressing.kicad_sch		
Title: Konami 007783		
Size: A3	Date: 2023-10-10	**Rev:**
KiCad E.D.A. 8.0.5		Id: 3/5



Test register 1



Sheet: /Misc/		
File: misc.kicad_sch		
Title: Konami 007783		
Size: A3	Date: 2023-10-10	**Rev:**
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Four time slots available all the time for addressing.
0: – OBJBUFINIT high: Writing zeros to all data in the Object Data Buffer.
– DMA high: Data copying from Object RAM to Object Buffer RAM.
1: Parsed object data copy to the frame buffers.
2: Master CPU access
3: Slave CPU access

OBFI	DMA	OH2	$\overline{OH2}$	OH4	$\overline{OH4}$
1	X	0	1	0	1
X	1	0	1	0	1

OA[12..0]
OPDA[12..0]
AB[13..1]
SAB[13..1]
ORA[12..0]
{HCNT}

C79 F38 ORA0 — OA0, OPDA0, AB1, SAB1
B70 D72 ORA1 — OA1, OPDA1, AB2, SAB2
C91 E57 ORA2 — OA2, OPDA2, AB3, SAB3
B58 G19 ORA3 — OA3, OPDA3, AB4, SAB4
B106 C70 ORA4 — OA4, OPDA4, AB5, SAB5
B97 D94 ORA5 — OA5, OPDA5, AB6, SAB6
B88 C65 ORA6 — OA6, OPDA6, AB7, SAB7
B79 D81 ORA7 — OA7, OPDA7, AB8, SAB8
C105 C62 ORA8 — OA8, OPDA8, AB9, SAB9
A86 A83 ORA9 — OA9, OPDA9, AB10, SAB10
A106 A14 ORA10 — OA10, VSS, AB11, SAB11
A96 A8 ORA11 — OA11, VDD, AB12, SAB12
A70 A55 ORA12 — OA12, VDD, AB13, SAB13
A58 E55 $\overline{OOE}$ — $\overline{OAOE}$, VSS, MREAD, SREAD

OBFI, DMA → E30 R2N → H2 → A25 N2B → $\overline{TES1D6}$ → A31 CLK K3B → $\overline{OH2}$
H2 → A35 N2P → TES1D6 → A45 CLK K4B → V1N B9 → OH2, $\overline{OH2}$
H4 → A38 N2B → TES1D6 → A42 CLK K4B → $\overline{OH4}$
H4 → A22 N2P → TES1D6 → A19 CLK K3B → E8 V1N → OH4, $\overline{OH4}$

CPU Read/Write logic

$\overline{SLWR}$ → A50 V1N → SLWR
$\overline{SUWR}$ → A52 V1N → SUWR
$\overline{MOCS}$ → A11 V1N → MOCS
$\overline{SOCS}$ → A16 V1N → SOCS

Object Character RAM Read/Write logic

SLWR, SAB1 → R87 R3B → $\overline{CRW}$0
SUWR, SAB1 → R58 R3B → $\overline{CRW}$1
A57 R2N: TIM3, $\overline{OCRA}$
SLWR → Q80 R3B → $\overline{CRW}$2
SUWR → Q58 R3B → $\overline{CRW}$3
SAB1 → Q84 V1N
$\overline{CRW}$[3..0]

Object RAM Read/Write logic

SOCS, SLWR, OH4, OH2 → B68 N4N
A6 V1N: $\overline{MLWR}$
MOCS, OH4, $\overline{OH2}$ → B2 N4N → B55 N2N → B50 CLK K3B: TIM
B50 → D40 R2N → $\overline{ORLWR}$

BEFLAG & DMA & DMAON
BFLGD, H8, H2 → E11 N8B
DMA, $\overline{OH4}$, $\overline{OH2}$ → E3 N2N → D4 CLK K3B
E20 FDG: DMAON, OBFI, CLR
VDD
D10 BD3: CLK2
OBFI, $\overline{OH4}$, $\overline{OH2}$ → E5 D24
OBFI & DMAON

SOCS, SUWR, OH4, OH2 → A68 N4N → B57 N2N → B46 CLK K3B: TIM → D36 R2N → $\overline{ORUWR}$
A3 V1N: $\overline{MUWR}$
MOCS, OH4, $\overline{OH2}$ → B5 N4N → B57

Sheet: /Address output/
File: address_output.kicad_sch

Title: Konami 007783

Size: A3	Date: 2023-10-10	**Rev:**
KiCad E.D.A. 8.0.5		Id: 6/5